



F1-38 Hele Shaw Apparatus

The Armfield F1-38 Hele Shaw apparatus enables investigation into the principles of potential flow and enables modelling of appropriate physical systems.

It allows students to study various source and sink arrangements and look at flow around a variety of different shaped models.

PRODUCT CODE: F1-38

Categories: Educational, F1 Series Fluid Mechanics, F1-38 Hele Shaw Apparatus, IFT Industrial Food Technology

Tags: Aerofoil Gottingen , Dye , F1 , F1-35 , F1-38a , F1-38b , Flow Patterns , Fluid Dynamics , Fluid Mechanics , Hele-Shaw , Potential Flow , Rankine , Sinks And Sources

DESCRIPTION

The F1-38 Hele Shaw Apparatus provides a very visual means of showing potential flow in the field of fluid dynamics. The water flows through a very narrow channel which is formed between two plates causing the water to flow in a special way.

The mean velocity of the water along the channel obeys equations as those for local velocity in potential flow. The water flow within the channel is designed to be in a steady, laminar state.

The apparatus injects dye (by gravity) into the water flow to produce streamlines that clearly show the effect of different flow conditions around shapes and/or with the use of sinks and sources. **F1-38 Hele Shaw Apparatus** is a demonstration of classic experiments with visualisation of flow behaviour by injection of dye as a steady flow in a working cell.

The apparatus is designed to fit onto the top of the standard F1-10 series hydraulic bench forming part of the F1 series fluid mechanics' range. The **F1-10 Hydraulics Bench** can be used as work surface and water collection sump.

The apparatus requires a supply of clean cold water, with an optional **overhead tank (F1-38a)** supplied separately, available if required. In addition the optional **F1-38b Visualisation accessory** allows the user to capture or project video or still images via use of the supplied USB camera.

[Click here to view video of F1-38](#)

TECHNICAL SPECIFICATIONS

Working section

Viewing Area:	892mm
Sinks/source/closed:	4 points
Dye injectors:	16

Dye: Food based dye
Flow control valve: Operates fully closed (0) to fully open(11)
Camera requirements: USB – PC: Windows 10, 8, 7, Vista, XP
Operational water pressure min/max: 0.2 – 0.3bar

FEATURES & BENEFITS

- Highly visual apparatus enabling the demonstration of a wide variety of flow patterns around different shapes
- Includes dye injection system
- Accessory to F1-10 bench
- Supplied with spare material for user defined shapes
- Optional visualiser F1-38b allows the user to capture or project video or still images via use of the supplied USB camera
- Supplied with comprehensive manual

Demonstration Capabilities & Video

Ideal flow around immersed bodies

1. Cylinder
2. Aerofoil
3. Bluff body

Ideal flow associated with sinks and sources

1. Formation of a Rankine half body
2. Formation of a Rankine oval
3. Circular streamlines from a doublet
4. Superposition of sinks and sources User defined shape

Ideal flow in channels and at boundaries

1. Convergent channel
2. Divergent channel
 1. 90-degree bend
 2. Sudden contraction
 3. Sudden enlargement

Models Supplied

A set of basic model shapes are provided to analyse flow patterns around these. (Further models can be cut from the same material used in for the standard models to ensure compatibility with the unit).

Models include:

Venturi half **X2**
Orifice plate half **X2**
Disc
Aerofoil NACA633-618
Aerofoil Gottingen 535
Blank material

Optional Extras

F1-38a Header Tank: If mains water cannot be connected

F1-38b Visualiser: Autofocusing. Still and video capture **Mac:** OS 10.12 and later (Mac OSX 10.5 and later compatible)

PC: Windows 10, 8, 7, Vista, XP

Ordering specification

- Desktop or F1-10 bench mounted apparatus, demonstrates ideal flow, and Hele Shaw principles and is able to accept student-made models
- Working section dimensions 345mm x 265mm
- Actual viewing area 295mm x 245mm
- Comprehensive instruction manual with illustrations

Requirements

- **Water supply**
- **Drain:** Suitable for water containing dye



Shipping Specifications

Packed and crated shipping specifications

Volume: 0.125m³

Gross weight: 10Kg

Dimensions

Length: 0.50m

Width: 0.44m

Height: 0.78m

Order Codes

- F1-38
- F1-38a
- F1-38b

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